Topic 9 — Transport & Communication

> **Sources baked in: Ghatnachakra** World Transport/Ports/Mapping (CB–247+); NCERT Class 10–12 transport, Drishti / PW heads, **Ghatnachakra** Transport (CA–352+) + **Tourism / Misc** (CA–376+), UPPCS Prelims 2018–2025 **Weight:** ★★★★ — zone–HQ, port–state, GQ / NS–EW / Jhansi, tourism match lists, scheme year, UP expressway/airport **Last verified:** August 2026 **Current Affairs:** Gati Shakti 2021; NLP 2022; New Pamban; FLY91; Jewar (Noida Intl); Amrit Bharat UP 149 stations

Current Affairs (this topic)

Year	Fact	Why it matters
13 Oct 2021	**PM Gati Shakti** National Master Plan launched (not 2022)	Year trap with seven engines
17 Sep 2022	**National Logistics Policy** — complements Gati Shakti	Scheme pair
Feb 2023	Amrit Bharat: about **1,275** stations all-India; **149** of them in UP for modernisation	149 = UP stations, not new lines
2024–26	**New Pamban** vertical-lift railway sea bridge (RVNL)	RVNL + first vertical-lift fact
18 Mar 2024	**FLY91** inaugural from **Manohar Intl** (Mopa, Goa (गोवा))	New airline + airport pair
2024–26	**Noida International (Jewar)** — UP's new greenfield intl hub	UP airport CA
DFC	Western Dadri (ददरी)–JNPT / Eastern Ludhiana–Dankuni — phased opening	Freight spine

Sagarmala / Bharatmala remain living umbrella programmes; still tests **name swap**, not project lists.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q88

Which of the following ports is on the western coastline?

A. Nhava Sheva B. Visakhapatnam C. Chennai D. Tuticorin

Show answer

Logic: Three eastern ports — one western settles it. **Ans: A.** Nhava Sheva (JNPT) near Mumbai is western; the other three are eastern.

Q. UKPCS Prelims 2025, Q92

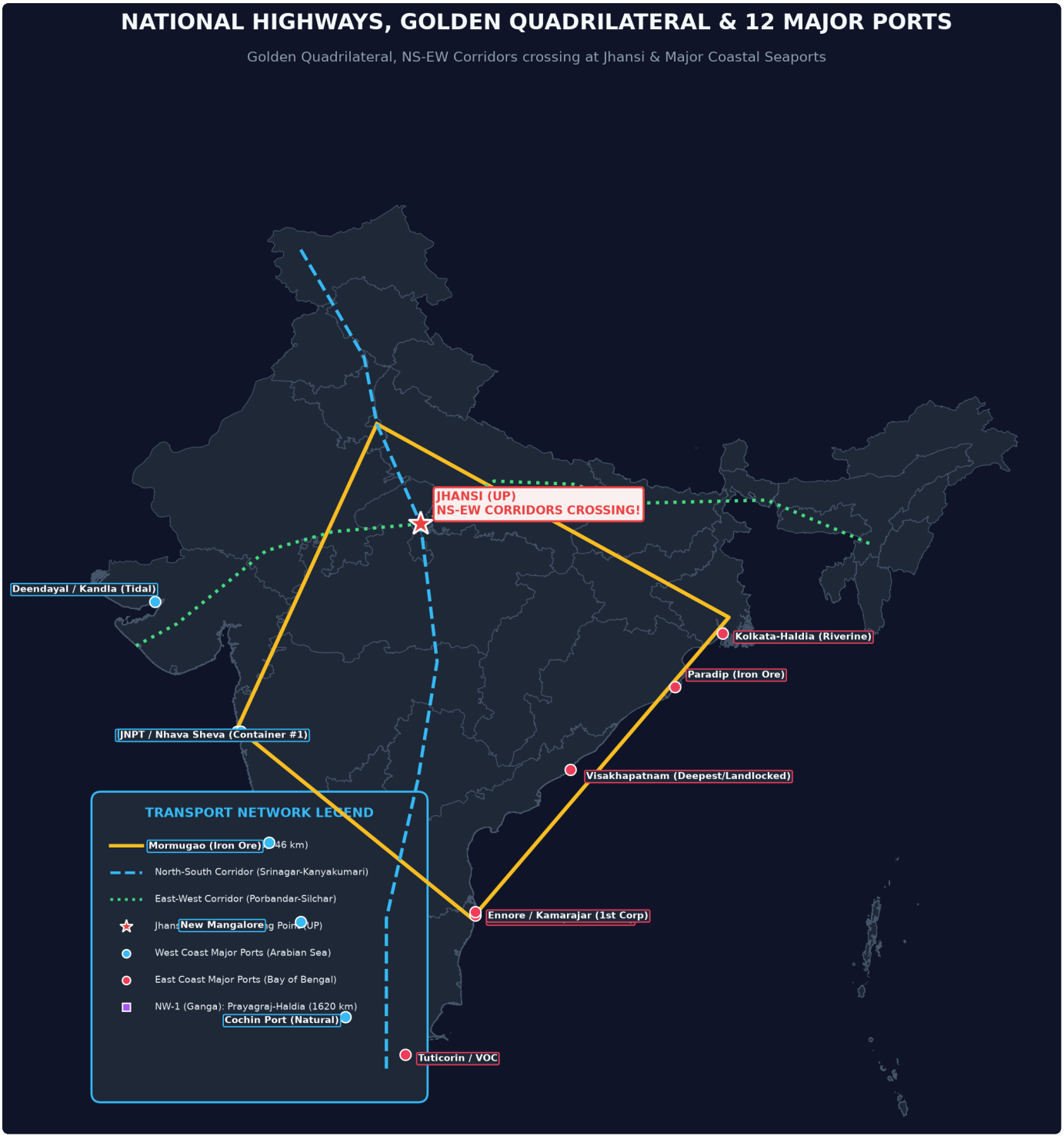
Which country has the highest density of railway network?

A. Canada B. U.S.A. C. Russia D. India

Show answer

Logic: Density is route length per area — Series B keys USA among the four (चातुर्याम). **Ans: B.** Among the given options, the answer is the **U.S.A.** India is a common distractor on railway-density stems.

N.0 Framework (map first)



Transport and communication are the **arteries of circulation** of the economy.

- **Transport** moves people and goods.
- **Communication** moves information, ideas and messages.
- Principal modes are **land** (road, rail), **water** (inland + sea), **air**, and **pipelines**.
- Roads often feed other modes: they link farms to stations, ports and airports.
- Rail suits bulky, long-distance freight and mass passenger movement on fixed tracks.
- Sea shipping dominates international bulk trade; inland waterways suit low-cost, low-speed cargo on navigable rivers and canals.

- Air suits high-value or time-sensitive cargo and long-distance passengers.
- Pipelines suit continuous flow of oil, gas and products once the capital line is laid.
- About **95% of India's trade by volume** (and a large share by value) moves by **sea**.
- Inland waterways still carry only a **tiny** share of cargo, so schemes try to raise that share.
- Multimodal planning under **PM Gati Shakti** and the **National Logistics Policy** aims to cut logistics cost and time.
- Map reading (रीडिंग) for this chapter pairs **routes** (GQ, DFC, NW-1, HBJ) with **nodes** (ports, airports, railway HQs and UP expressway towns).

N.1 Railways

Indian Railways is the country's principal long-distance land carrier for passengers and bulky freight.

- The first passenger train ran from **Mumbai to Thane in 1853** (about 34 km).
- Gauge widths are **broad gauge** about **1.676 m (5½ ft)**, metre gauge **1 m**, and narrow gauge **762 / 610 mm**.
- Most traffic is now broad gauge.
- Indian Railways is often described as the **largest public-sector undertaking** in the country.
- The network is organised into **railway zones**, each with a fixed headquarters. Count in current notes is **18 zones** (including South Coast Railway at **Visakhapatnam**). Older books still say 16 — read the stem.
- **Konkan Railway** runs **Roha (Maharashtra)–Mangaluru (Karnataka)** through Maharashtra–Goa–Karnataka. It does **not** enter Kerala as a Konkan Railway main line.
- **Pir Panjal** rail tunnel (~**11.2 km**) is the classic long Himalayan rail tunnel.
- **Fairy Queen** uses one of the world's oldest working steam engines (1855 / re-launch).

Factory (फैक्टरी) / works	Place	What it builds
Rail Coach Factory	**Kapurthala**	Coaches
Integral Coach Factory	**Perambur** (Chennai)	Coaches
Chittaranjan Locomotive Works	**Chittaranjan**	Electric locomotives
Banaras Locomotive Works	**Varanasi (वाराणसी) / Manduadih**	Diesel–electric locomotives
Modern Coach Factory	**Rae Bareli**	Third major coach factory

- **National Academy of Indian Railways** (old Railway Staff College) is at **Vadodara**.
- First **CNG** passenger train: **Rewari–Rohtak**. First PPP rail line note: **Gandhidham–Tuna Tekra (Gujarat)**.
- **Darjeeling Himalayan Railway** (Siliguri–Darjeeling) is a UNESCO Mountain Railway of India site (with Nilgiri (नीलगिरि) and Kalka–Shimla).
- **Mumbai–Ahmedabad (अहमदाबाद)** is India's first high-speed / bullet-train corridor project (~508 km; Japan partnership).

- Sal / sakhu hard wood is the classic **railway sleeper** timber fact.

Zone	Headquarters
Northern	New Delhi
North Eastern	**Gorakhpur**
North Central	Prayagraj
East Central	**Hajipur**
Eastern	Kolkata (Fairlie Place)
South Eastern	**Garden Reach, Kolkata**
Southern	Chennai
South Central	**Secunderabad** (not Hyderabad (हैदराबाद) city trap)
Western	Churchgate, Mumbai
West Central	Jabalpur
Central	Mumbai CSMT
East Coast	Bhubaneswar
North Western	**Jaipur** (not Jodhpur)
South Western	**Hubballi**
South East Central	Bilaspur
South Coast	Visakhapatnam
Metro Railway Kolkata	17th zone (2010)

Dedicated Freight Corridors

- The **Western DFC** links **Dadri** to **JNPT**.
- The **Eastern DFC** links **Ludhiana** to **Dankuni**.
- DFCs move bulk freight off crowded passenger tracks so industry gets faster rail haulage.

Amrit Bharat Station Scheme

Amrit Bharat redevelops **existing passenger stations**. It upgrades buildings, platforms, circulating area, accessibility and passenger amenities.

It does **not** build new railway lines or new tracks. The number **149** is not a count of new lines.

As of **February 2023**, the Ministry of Railways had identified about **1,275** stations across India under the scheme. **Uttar Pradesh's share (भाग) of that list was 149 stations.**

Inline PYQ — UPPCS Prelims 2023, Q71

How many railway stations have been identified for modernization under the Amrit Bharat Station Scheme in Uttar Pradesh as on February 2023?

A. 57

B. 82

C. 123

D. 149

Show answer

Ans: D — 149 is the number of **UP stations** identified for modernisation, not 149 new railway lines.

New Pamban and Hubballi museum

- The New Pamban bridge toward Rameswaram was built by **Rail Vikas Nigam Limited (RVNL)**.
- It is India's **first vertical-lift railway sea bridge**.
- A railway museum was dedicated in **August 2020** at **Hubballi**, which is also the South Western Railway headquarters.

Inline PYQ — UPPCS Prelims 2025, Q136

With reference to the New Pamban Bridge, which of the following statements is/are correct?

1. It has been constructed by Rail Vikas Nigam Limited, a Navratna PSU under the Ministry of Railways.
2. It is India's first vertical lift railway sea bridge.

Select the correct answer from the code given below:

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: C — Both statements are correct.

Inline PYQ — UPPCS Prelims 2020, Q31

In August 2020, Railway Museum in India was dedicated to the nation at which of the following places?

A. Hubballi

B. Mysore (मैसूर)

C. Silvasa

D. Chittaranjan

Show answer

Ans: A — Hubballi.

Inline PYQ — UPPCS Prelims 2025, Q150

With reference to Indian Railways, which of the following pairs is/are correctly matched?

1. Northern Railway — New Delhi
2. North Eastern Railway — Gorakhpur
3. South Eastern Railway — Cuttack

Select the correct answer from the code given below:

A. Only 1 and 2

B. Only 2 and 3

C. Only 1 and 3

D. 1, 2 and 3

Show answer

Ans: A — Pair 3 is wrong. South Eastern Railway headquarters is at **Garden Reach, Kolkata**, not Cuttack.

N.2 Roads, Expressways, Bharatmala

Roads are preferred for short and medium hauls because they are cheaper to build than railways and act as feeders to rail, air and sea terminals.

Road classes

Class	Who builds / maintains	Note
National Highways	Central government / **NHAI**	About **2%** of road length but about **40%** of road traffic
State Highways	State governments	Link the state capital to district headquarters
District roads	Zila Parishad (जिला परिषद) / district	Link the district headquarters to important nodes
Rural roads	Local bodies; **PMGSY** push	About **80%** of total road length; all-season village links
Border roads	**BRO** (established May 1960)	Northern and north-eastern borders; high-altitude strategic infrastructure; executes **Project Dantak** in Bhutan, and infrastructure projects in Myanmar and Afghanistan
International highways	Cross-border links	Neighbour connectivity

- **NHAI** was operationalised in **1995** under the Ministry of Road Transport and Highways.
- The **Nagpur Plan (1943)** sorted Indian roads into National Highways, State Highways, District Roads and Village Roads — still the syllabus skeleton.
- India's road length is about **63+ lakh km** — the world's second-largest road network. Roads carry roughly **~90%** of passenger traffic and **~65–70%** of freight in recent NTDPC / MoRTH data (older figures used **~80%** passenger).
- National Highways are only about **2%** of road length but carry about **40%** of road traffic.
- National Highway length by state (NHAI, Dec 2022): **Maharashtra > Uttar Pradesh > Rajasthan > Madhya Pradesh**. Older data often ranked **Uttar Pradesh** (उत्तर प्रदेश) first.
- **State Highway** length leaders: **Maharashtra**, then Karnataka, Gujarat, Rajasthan (Basic Road Statistics).
- **Longest NH:** older data cited old **NH-7**; current MoRTH data often cite **NH-27 (EW)** Porbandar–Silchar. **NH-44** is also cited as the longest single corridor (Srinagar–Kanyakumari).
- **Golden Quadrilateral** links **Delhi–Mumbai–Chennai–Kolkata** (about **5,846 km**); launched under NHDP in **2001**.
- The **North–South Corridor** links **Srinagar** to **Kanyakumari** (कन्याकुमारी).
- The **East–West Corridor** links **Porbandar** to **Silchar** (not Dwarka (द्वारका) / Dibrugarh–Surat).
- **NS and EW corridors cross at Jhansi (Uttar Pradesh).**
- **Pradhan Mantri (मंत्री) Bharat Jodo Pariyojana** builds highway corridors to tourist and economic nodes on **BOT**.
- **Mumbai–Pune (पुणे)** was India's first six-lane access-controlled expressway (~95 km).
- **PMGSY** began **25 December 2000** — all-weather rural connectivity to eligible habitations.
- The **Jawahar (Banihal) Tunnel** crosses the **Banihal** axis of the **Pir Panjal** in Jammu & Kashmir (BRO).
- The **India–Myanmar–Thailand Trilateral Highway** is a road link from India through Myanmar into Thailand. It ends at **Mae Sot** in Thailand, not Chiang Mai.
- The **BCIM** corridor (Bangladesh–China–India–Myanmar) runs **Kunming to Kolkata**. It is not a Varanasi–Kunming highway.
- **Bharatmala** is the umbrella **highway** programme. It is **not** a port scheme.
- **Bharatmala Phase-I progress (MoRTH, as on 31 October 2024):** about **26,425 km** awarded and **18,714 km** constructed nationwide. Among states, **Rajasthan** had the **highest constructed length** (**2,241 km**), ahead of Uttar Pradesh and Maharashtra on that date.
- An **expressway** is an **access-controlled** high-speed road, stricter than an ordinary national highway.

Atal Tunnel

- The **Atal Tunnel** lies in the **Pir Panjal** range.
- It was built with **Border Roads Organisation** association.
- It connects Manali toward the Lahaul–Spiti side (Rohtang axis).
- Do not mark a bare claim that it is "the world's longest highway tunnel" as true. The safe fact is **Pir Panjal**.

Inline PYQ — UPPCS Prelims 2025, Q44

With reference to the Atal Tunnel, which of the following statements is/are correct?

1. This tunnel is the world's longest highway tunnel.
2. This tunnel is built in the Pir Panjal range of the Himalayas.

Select the correct answer from the code given below:

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: A — Only statement 2 is correct. The world's-longest claim is not accepted.

Purvanchal Expressway

The **built** Purvanchal Expressway runs Lucknow–Barabanki–Ambedkar (अम्बेडकर) Nagar–Azamgarh–**Mau**–Ghazipur–Ballia.

- It **does pass Mau**.
- It does **not** run through **Basti** on the main alignment.

On the **2018** Prelims, the proposal-stage "will not pass" question's official answer was **Mau**. On today's map, Mau lies on that corridor.

Inline PYQ — UPPCS Prelims 2018, Q3

'Purvanchal Express Way' in Uttar Pradesh will not pass through which of the following cities?

A. Faizabad

B. Basti

C. Azamgarh

D. Mau

Show answer

Ans: D — This is the **2018** key for that wording. The **built** expressway **does** pass Mau. Basti is not on the main alignment.

Expressway	Note
Yamuna (यमुना)	Greater Noida–Agra
Agra–Lucknow	West-central Uttar Pradesh
Purvanchal	Lucknow–Ballia via Azamgarh and **Mau**
Bundelkhand	Southern Uttar Pradesh

Expressway	Note
Ganga Expressway	Meerut (मेरठ)–Prayagraj corridor (under construction / phased)
Gorakhpur Link	Connector toward Gorakhpur / NER city

N.3 Waterways

Inland and coastal water transport is cheap for bulk cargo when channel depth and terminals exist.

- The **Inland Waterways Authority of India (IWAI)** was set up in **1986**.
- The **National Waterways Act, 2016** declared **111** National Waterways (the older five plus many new stretches).
- Inland waterways still move only a **tiny** share of Indian cargo (under about 1% in older notes). Sea transport still carries about **95%** of trade by volume.

Learn the first five National Waterways by river, stretch and state. They are the standard map, not an optional extra.

NW	River / canal	Stretch	States / note
NW-1	Ganga–Bhagirathi (भागीरथी)–Hooghly (हुगली)	**Haldia–Prayagraj** (about 1,620 km)	UP, Bihar, Jharkhand, West Bengal
NW-2	Brahmaputra	**Dhubri–Sadiya**	Assam
NW-3	West Coast Canal, Champakara and Udyogmandal canals	**Kottapuram–Kollam**	**Kerala**
NW-4	Godavari, Krishna and connected canals (including Buckingham Canal)	**Kakinada–Puducherry**	Andhra Pradesh–Tamil Nadu–Puducherry
NW-5	Brahmani–Mahanadi delta (डेल्टा) and East Coast Canal	**Talcher–Dhamra** / Geonkhali–Charbatia stretch	Odisha–West Bengal

- Uttar Pradesh sits on **NW-1**.
- **Jal Marg Vikas** is the capacity-upgrade project on **NW-1**, with multimodal terminals such as **Varanasi** (वाराणसी), Sahibganj and Haldia.
- The **Barak** (बराक) stretch in Assam (Lakhipur–Bhanga) is another named National Waterway in the 2016 list; do not treat it as NW-1 or NW-2.

In **July 2020**, the first containership movement from **Kolkata Port to Agartala** used the **Chattogram (Chittagong)** route, not Benapole, Maloti or Payra.

Inline PYQ — UPPCS Prelims 2020, Q71

First ever containership from Kolkata Port to Agartala was sent through which one of the following Ports of Bangladesh in July 2020?

A. Benapole

B. Maloti

C. Payra

D. Chattogram (Chittagong)

Show answer

Ans: D — Chattogram, not Benapole, Maloti or Payra.

N.4 Ports, Harbours, Maritime Trade

India's mainland plus island coasts stretch about **7,517 km**.

- A **harbour** is sheltered water where ships (नावाध्यक्ष) can ride safely.
- A **port** is a harbour **plus** docks, cargo gear and passenger facilities.
- A **natural harbour** already has coastal shelter; an **artificial harbour** needs breakwaters and dredging.
- Mumbai, Cochin and Visakhapatnam are often cited as **natural**; **Chennai** is often cited as **artificial** / open seaport.
- **Kolkata–Haldia** is a major **riverine** port complex; **Haldia** is the classic **outport** of Kolkata.
- **Kandla (Deendayal)** on the **Gulf of Kutch** is a **tidal** natural harbour; **Diamond Harbour** is also tidal harbour.
- **JNPT / JNPA (Nhava Sheva)** was built to **decongest Mumbai** and is India's container flagship.
- **Cochin Shipyard** is India's largest shipyard fact; **Alang (Gujarat)** is the largest ship-breaking yard.
- **Paradip** was developed to ease congestion on **Kolkata and Visakhapatnam**; iron-ore export to Japan.
- **Sethusamudram** aims to link the **Gulf of Mannar** with **Palk Bay** / Palk Strait for a continuous peninsula route.
- **LNG terminals**: Dahej, Hazira, Kochi, Dabhol, Ennore — **not Kandla**.
- Kochi sits on the **Malabar Coast**.
- India has **12 major ports** and about **200** non-major ports (**Vadhavan, Maharashtra** is the 13th major port project).
- Roughly **95% of trade by volume** moves by maritime transport.
- **Uttar Pradesh has no major seaport.**
- **Kandla (Deendayal)** serves a large north-western hinterland, including petroleum products.
- **Mormugao** is classically an **iron-ore** export port.
- **Paradip** handles iron ore and coal on the Odisha coast.
- **Ennore (Kamarajar)** is a major port on the Tamil Nadu coast, north of Chennai — first major port registered as a company (कंपनी); **Mundra** is the largest privately owned port.

Coast	Major ports
West / NW	Kandla (**Deendayal**, Gujarat), Mumbai, JNPT, Mormugao (Goa), New Mangalore (Karnataka), Cochin (Kerala)
East / SE	Kolkata–Haldia, Paradip (Odisha), Visakhapatnam (Andhra Pradesh), Chennai, Ennore (**Kamarajar**), Tuticorin (**VOC**, Tamil Nadu)

Port–state match bank (next year can ask any)

2025 Q50 mixed **major** and **non-major** names. Learn the bank. ★ marks that paper's set.

Port	State	Tag
Kandla / Deendayal	**Gujarat**	Major; NW hinterland
Mumbai	**Maharashtra**	Major; natural harbour class
JNPT / JNPA	**Maharashtra**	Container flagship
Mormugao	**Goa**	Major; iron-ore note
New Mangalore	**Karnataka**	Major
Cochin / Kochi	**Kerala**	Major; natural harbour class
Alappuzha ★	**Kerala**	Non-major; 2025 Q50
Kolkata–Haldia	**West Bengal**	Major; riverine
Paradip ★	**Odisha** (ओडिशा)	Major; 2025 Q50
Visakhapatnam	**Andhra Pradesh**	Major; natural harbour class
Kakinada ★	**Andhra Pradesh**	Non-major; 2025 Q50
Krishnapatnam	**Andhra Pradesh**	Non-major / private class
Chennai	**Tamil Nadu**	Major; often artificial harbour
Ennore / Kamarajar	**Tamil Nadu**	Major; N of Chennai
Tuticorin / VOC ★	**Tamil Nadu**	Major; 2025 Q50
Mundra / Pipavav	**Gujarat**	Non-major / private west coast
Dhamra	**Odisha**	Non-major east coast

- **Uttar Pradesh has no major seaport.** Do not invent one in match lists.

Inline PYQ — UPPCS Prelims 2025, Q50

Match List-I with List-II and choose the correct answer using the code given below the lists.

List-I (Port)	List-II (State)
A. Paradip	1. Tamil Nadu

List-I (Port)	List-II (State)
B. Tuticorin	2. Kerala
C. Kakinada	3. Odisha
D. Alappuzha	4. Andhra Pradesh

> Row order is **not** the answer — use the option codes.

A. 1 3 2 4

B. 3 1 2 4

C. 3 1 4 2

D. 4 1 3 2

Show answer

Ans: C — Paradip–Odisha; Tuticorin–Tamil Nadu; Kakinada–Andhra Pradesh; Alappuzha–Kerala.

Inline PYQ — UPPCS Prelims 2023, Q60

Which of the following (Port - Country) is not correctly matched?

A. Rotterdam - Netherlands

B. Igarka - China

C. Montevideo - Uruguay

D. Jakarta - Indonesia

Show answer

Ans: B — Igarka is in **Russia**, not China.

N.5 Airports

Air transport is fast but costly, so it dominates high-value and time-sensitive flows plus long passenger hauls.

- **UDAN** is the regional connectivity scheme that tries to make flying viable to smaller cities.
- India's first airmail / air transport flight took place in **1911** between **Prayagraj (Allahabad) and Naini** (world's first commercial airmail flight by Henri Pequet).
- **Indian National Airways Company** was established in **1933**.
- **Nationalisation of Civil Aviation (1953)**: All airline companies were nationalised and merged under two corporations: **Air India** (for international routes) and **Indian Airlines** (for domestic and regional routes to neighbours like Nepal, Bangladesh, Pakistan, Sri Lanka, Myanmar, Afghanistan, and

Maldives). **Vayudoot** was created in 1981 for regional feeder services and later merged with Indian Airlines.

- In 2007, Air India and Indian Airlines were merged under **National Aviation Company of India Limited (NACIL)**.
- Metro international hubs include Delhi (IGI), Mumbai, Chennai, Kolkata, Bengaluru, Hyderabad and Cochin.
- **Cochin** is India's first **solar-powered** airport (and the world's first fully solar-powered international airport).
- **Cochin** was India's first airport developed under a **public limited / PPP** model.
- **Kushinagar** airport in Uttar Pradesh got international status on **24 June 2020**.
- **Jewar** is **Noida International Airport**, Uttar Pradesh's new greenfield international hub.
- FLY91 is a newly launched Indian airline. Its inaugural flight on **18 March 2024** took off from **Manohar International Airport** (Mopa, Goa).
- Amritsar (अमृतसर)'s international airport is **Sri Guru (गुरु) Ram Das Jee** (Raja (राजा) Sansi).

Major Named Airports in India (Match Desk)

Airport Name	City / Location	State / UT
Chhatrapati Shivaji Maharaj International Airport	Mumbai (Santa Cruz / Sahar)	Maharashtra
Netaji Subhash Chandra Bose International Airport	Kolkata (Dum Dum)	West Bengal
Indira Gandhi International Airport	New Delhi (Palam)	Delhi NCT
Anna / Chennai International Airport	Chennai (Meenambakkam)	Tamil Nadu
Kempegowda International Airport	Bengaluru (Devanahalli)	Karnataka
Rajiv Gandhi International Airport	Hyderabad (Shamshabad)	Telangana
Sri Guru Ram Dass Jee International Airport	Amritsar (Rajasansi)	Punjab
Dr. Babasaheb Ambedkar International Airport	Nagpur (Sonegaon)	Maharashtra
Lokpriya Gopinath Bordoloi International Airport	Guwahati (Borjhar)	Assam
Veer Savarkar International Airport	Port Blair	Andaman & Nicobar
Sardar Vallabhbhai Patel International Airport	Ahmedabad	Gujarat
Chaudhary Charan Singh International Airport	Lucknow (Amausi)	Uttar Pradesh
Lal Bahadur Shastri International Airport	Varanasi (Babatpur)	Uttar Pradesh
Lok Nayak Jayaprakash Airport	Patna	Bihar
Devi Ahilyabai Holkar Airport	Indore	Madhya Pradesh
Raja Bhoj Airport	Bhopal	Madhya Pradesh

Inline PYQ — UPPCS Prelims 2022, Q27

The first airport powered by solar energy in India is-

A. Chennai

B. Cochin

C. Ahmedabad

D. New Delhi

Show answer

Ans: B — Cochin, not Delhi, Chennai or Ahmedabad.

Inline PYQ — UPPCS Prelims 2020, Q61

Which of the following Airports in Uttar Pradesh has been granted International Status on 24th June, 2020 by Union Cabinet of India?

A. Agra Airport

B. Prayagraj Airport

C. Gorakhpur Airport

D. Kushinagar Airport

Show answer

Ans: D — Kushinagar.

Inline PYQ — UPPCS Prelims 2024, Q90

With reference to "FLY91", which of the following statements is/are correct?

1. It is a newly launched airline in India.
2. Its inaugural flight took off from Manohar International Airport on 18th March, 2024.

Select the correct answer from the code given below:

A. Only 2

B. Neither 1 nor 2

C. Only 1

D. Both 1 and 2

Show answer

Ans: D — Both statements are correct.

N.6 Pipelines

Pipelines move oil, gas and products continuously after a high upfront capital cost.

- Once laid, running cost is low, and there is little trans-shipment loss compared with road or rail tanker haulage.
- Pipelines are best for liquids and gases that must flow steadily from fields and terminals to refineries, fertilizer plants and power stations.
- **GAIL** (set up **1984**) transports, processes and markets natural gas on the national grid.
- The **Hazira–Bijaipur–Jagdishpur (HBJ / HVJ)** gas pipeline is the classic cross-country gas trunk toward western and north-central markets.
- HBJ draws gas linked to the **Mumbai High / Bassein** offshore fields and feeds fertilizer and power complexes along the Gujarat–Madhya Pradesh–Uttar Pradesh belt, including nodes near **Auraiya** and **Jagdishpur** in Uttar Pradesh.
- The oldest major crude network taught in school maps runs from **Naharkatiya** (Assam) through **Noonmati / Guwahati** to **Barauni**, with later links toward Kanpur (कानपुर) and Haldia branches.
- The **Salaya–Koyali–Mathura** crude line brings imported west-coast crude inland to Gujarat and **Mathura** (मथुरा), with further northern extensions toward Panipat (पानीपत) and Jalandhar in standard notes.
- Domestic refinery product lines include feeds around **Mathura** serving north Indian markets.
- In **September 2019**, the product pipeline from **Motihari (India)** to **Amlekhganj (Nepal)** was inaugurated. It is not Motihari–Kathmandu and not Darbhanga–Amlekhganj.

Inline PYQ — UPPCS Prelims 2019, Q34

Cross border oil pipeline inaugurated by P.M. Modi in September 2019 connects which of the following cities?

A. Motihari and Amlekhganj

B. Darbhanga and Amlekhganj

C. Motihari and Kathmandu

D. None of these

Show answer

Ans: A — Motihari (Bihar) to **Amlekhganj** (Nepal).

N.7 Sagarmala, Blue Economy, Gati Shakti, NLP

Blue Economy — sector teaching card

- **Blue Economy** uses ocean and coastal resources for sustainable growth — fisheries, ports, shipping, offshore energy, coastal tourism, and marine biotechnology.
- India's **Sagarmala** focuses on port modernisation and coastal connectivity.
- **PM Gati Shakti** integrates infrastructure planning (roads, rail, ports, waterways) on one digital platform.
- **National Logistics Policy (2022)** targets lower logistics cost and multimodal freight efficiency.

Scheme	Focus	Trap
Sagarmala	Port-led development, coastal shipping, port connectivity, coastal communities	Not a highway scheme
Bharatmala	Highway corridors and feeders	Not ports
PM Gati Shakti NMP	Multimodal planning map of projects	Year trap
National Logistics Policy	Lower logistics cost and time	Complements Gati Shakti; does not replace it
Blue Economy	Sustainable ocean-resource use — shipping, fisheries, coastal tourism, offshore energy	Not “any coastal factory”
DFC	Dedicated rail freight hardware	Neighbour of Sagarmala, not the same scheme

- **PM Gati Shakti** National Master Plan was launched on **13 October 2021**, not in 2022.
- Its **seven engines** are roads, railways, airports, ports, mass transport, waterways and logistics.
- **National Logistics Policy** (September **2022**) works **with** Gati Shakti to cut logistics friction.

Inline PYQ — UPPCS Prelims 2023, Q136

With reference to PM Gati Shakti, which of the following statements is/are correct?

1. It was launched in 2022.
2. It has seven engines including roads, railways, airports, ports, mass transport, waterways and logistics.

Select the correct answer from the code given below:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Ans: B — Only statement 2 is correct. The launch year is **2021**, not 2022.

N.8 Communication (Basic)

- Transport carries **people and goods**.
- Communication carries **information**, ideas and messages.
- **Personal** communication includes post, telephone and internet messaging.
- **Mass** communication includes newspapers, radio, television and cinema.

- **AIR** and **Doordarshan** are the classic public broadcast arms.
- **BharatNet** is the rural broadband push.
- Indian communication satellites (INSAT / GSAT family) serve remote links, TV and weather.
- Optical fibre guides light through the core by **total internal reflection** at the core–cladding boundary.
- Do not mark diffraction, scattering, or ordinary refraction as the fibre fact.
- An optical-fibre cable is **not** an oil pipeline.

Inline PYQ — UPPCS Prelims 2019, Q125

The phenomenon used in optical fibre for transmission of light energy is

A. Total internal reflection

B. Diffraction

C. Scattering

D. Refraction

Show answer

Ans: A — Total internal reflection, not ordinary refraction.

N.9 Uttar Pradesh Focus

Uttar Pradesh has **no major seaport**. Its identity is rail headquarters, expressways, airports, NW-1, inland pipeline nodes and station redevelopment.

- **North Eastern Railway** has its headquarters at **Gorakhpur**.
- **North Central Railway** is headquartered at **Prayagraj** (प्रयागराज), so western and central UP sit on a major rail command map as well.
- The built **Purvanchal Expressway** spine is Lucknow–Azamgarh–**Mau**–Ghazipur–Ballia. It **passes Mau**; Basti is off the main alignment.
- **Kushinagar** became an international airport on **24 June 2020**.
- **Jewar** is the Noida International Airport project serving the western UP / NCR side.
- Under **Amrit Bharat**, about **149** UP stations were identified for modernisation as of **February 2023**, out of roughly **1,275** stations all-India.
- **National Waterway-1** on the Ganga runs **Haldia–Prayagraj**, so eastern and southern UP link to the inland waterway trunk.
- Western UP / NCR feel the pull of the **Western Dedicated Freight Corridor** toward the **Dadri–JNPT** freight spine.
- Pipeline geography places UP on the receiving end of HBJ gas toward **Jagdishpur / Auraiya** and of west-coast crude / product feeds centered on **Mathura**.

Theme	Fact
Railway HQ	**North Eastern Railway — Gorakhpur**
Expressway	Built Purvanchal **passes Mau**; Basti is off the main alignment
Airport	Kushinagar international from **24 Jun 2020**; Jewar = Noida International
Stations	Amrit Bharat: **149 UP stations** identified for **modernisation** (Feb 2023), out of about 1,275 all-India
Waterway	**NW-1** on the Ganga (Haldia–Prayagraj)
DFC pull	Western UP / NCR toward the Dadri–JNPT freight spine
Pipeline nodes	HBJ gas toward Jagdishpur / Auraiya; Mathura refinery product / crude complex

N.10 Tourism geography

Tourism sells India's **unity in diversity**. The Ministry's brand is **Incredible India**. **ITDC** was set up in **October 1966**. Niche tourism pushes India as a **365-day** destination for specialised interest travel. Visa-on-arrival / e-tourist visa raised tourist arrivals.

Resort / hill-station match bank

Place	State / UT	Note
Chakrata	**Uttarakhand** (उत्तराखंड) (Dehradun (देहरादून))	Cantonment; Tiger Fall
Haflong	**Assam**	Only hill station of Assam (Dima (डिम) Hasao)
Kalimpong	**West Bengal**	Darjeeling hills
Kufri	**Himachal Pradesh**	Near Shimla
Kasauli	**Himachal Pradesh**	Solan
Keylong	**Himachal Pradesh**	Lahaul
Auli (औली)	**Uttarakhand**	Ski / hill station
Ooty (Udhagamandalam)	**Tamil Nadu**	Nilgiris
Gulmarg / Pahalgam	**J&K**	Valley hill stations
Chikmagalur	**Karnataka**	Coffee / hills
Khajjiar	**Himachal Pradesh**	Meadow / lake note
Banni grassland	**Gujarat**	Kachchh

Place	State / UT	Note
Bugyal	**Uttarakhand**	Alpine meadow

City nicknames and pilgrimage geography

Fact	Correct	Trap
Pink City	**Jaipur**	Not Mysore
City of Lakes	**Udaipur**	**Not Jaisalmer**
City of Joy	Kolkata	Common nickname pair
City of Mahakal	Ujjain (उज्जैन)	Correct with Jaipur pair
Incredible India	MoT brand	Not “Beautiful India”
Sabarimala	**Kerala** (Ayyappa)	Not TN / KA / AP
Srisailam	**Nallamala Hills**, Andhra Pradesh	Mallikarjuna + Bhramaramba; Jyotirlinga
Omkareshwar	Mandhata island, **Narmada** (नर्मदा), Madhya Pradesh	**Not** Satmala Hills
Pushkar	Ajmer (अजमेर), **Aravalli** (अरावली) / Brahma temple	**Not** Mahadeo Hills
Biligiriranga Hills	SE Karnataka	Standing **Ranganatha / Venkatesha** temple
Tirupati Venkateswara	**Seshachalam** (Andhra Pradesh)	Not Shevaroy / Jawadhu / Biligiri as the main Tirupati hill
Udvada	Gujarat	Zoroastrian fire temple
Point Calimere	Nagapattinam, Tamil Nadu	Apex of **Coromandel Coast**
Konark Sun Temple	Odisha coast; UNESCO **1984**; Narasimha Deva I	World Heritage
Champaner–Pavagarh	Panchmahal, Gujarat; UNESCO **2004**	World Heritage
Mamallapuram	Pallava (पल्लव) stone town, Tamil Nadu; UNESCO	World Heritage
CST Mumbai	Victorian Gothic terminus; UNESCO	World Heritage
Ghost towns	Kuldhara, Dhanushkodi, Lakhpat	**Charkhari** is **not** a ghost town (Mahoba, UP)

- IAS **2005**: Champaner–Pavagarh, CST Mumbai, Mamallapuram and Konark are **all** on the UNESCO World Heritage List.
- IAS **2015** pilgrimage pairs: only **Srisailam–Nallamala** is correct among the three given.

N.12 World transport · ports · isopleths

Sea / air / rail pairs

Note	Fact
Busiest ocean trade route	**North Atlantic** (links two industrial zones)
Trans-Siberian	Long Eurasian rail; **Sochi** is **not** on the main route
Eurotunnel / Channel Tunnel	**England–France**
Canadian Pacific	Montreal–Vancouver
Bullet / HSR network length	**China** leader
Rotterdam	Rhine mouth; Europe's great seaport
Shanghai	Often #1 container harbour in recent frames
Santos	**Coffee port** (Brazil)
Maracaibo	Venezuela **oil** port
Port Nolloth	South Africa **diamond** port
Gwadar	**Pakistan** (Balochistan), Arabian Sea
Igarka	**Russia** (Yenisei) — not China
Alexandria	**Egypt** (मिस्र)

Isopleths / relief

Line	What it joins (equal value)
Isobar	Pressure
Isohyet	Rainfall
Isotherm	Temperature
Isohaline	Salinity
Isobath	Depth
Isogonic	Magnetic declination
Contours	Elevation / relief

Daily weather isobar maps are **isopleth** maps. Magellan's voyage is the first circumnavigation.